Short summary/Objective: Develop geographically accurate Radar simulation; characterize radar strength with standard corner reflector; build HDR algorithms; and implement a self-labelling object tracking algorithm

Summary: Using all new data, the HDR system proposed in the July report was tested and implemented; Utilizing Kalman tracking from team, a self-labelled object tracking algorithm was implemented; and implementing the new T-MAE model, a novel object tracking algorithm was implemented

Sea-ing in Binary: Extracting Object Locations and Higher Fidelity From Radar

Part 1: Overview of Outcomes

Accurate maritime perception from a compact X-band radar demands reconciling low-dynamic-range, noise-laden echoes with the precision required for collision avoidance, search-and-rescue, and autonomous navigation. Over the reporting interval, four parallel efforts addressed this gap.

Corner-reflector measurements captured the radar's native response curve and translated it into a stochastic signal model. A High-Dynamic-Range pipeline fused multi-gain sweeps into single 16-bit PPI frames, raising peak signal-to-clutter ratio by more than twelve decibels. A Monte-Carlo radar simulator ingests GIS land masks and buoy databases to generate pixel-level ground truth, enabling controlled algorithm training and failure-case rehearsal. Finally, two tracking architectures reached functional status: a self-initializing Kalman filter that learns background clutter on the fly, and a Temporal Masked Auto-Encoder trained from only three human-labelled frames, establishing both a baseline and a long-term learning roadmap. Together these efforts close the loop from raw RF data to labelled tracks and provide a reproducible platform for continued data collection in varied weather conditions.

Part 2: Radar Characterization and Data collection

Field measurements began on open terrain that provided an unobstructed path of roughly one hundred to one hundred fifty meters. A lightweight configuration interface, exposed through a single text file, allowed every primary radar parameter to be scripted, eliminating manual knob-turning between trials. Corner reflectors were fixed at two surveyed positions, and gain was swept from zero to full scale. Returns reached saturation at a gain setting of nine (on a zero-to-one-hundred scale); further increases produced no measurable rise in peak intensity, indicating that the receiver had entered compression. Figure 1 maps the reflector placements, while Figure 2 overlays measured signal strength on satellite imagery of the same site.

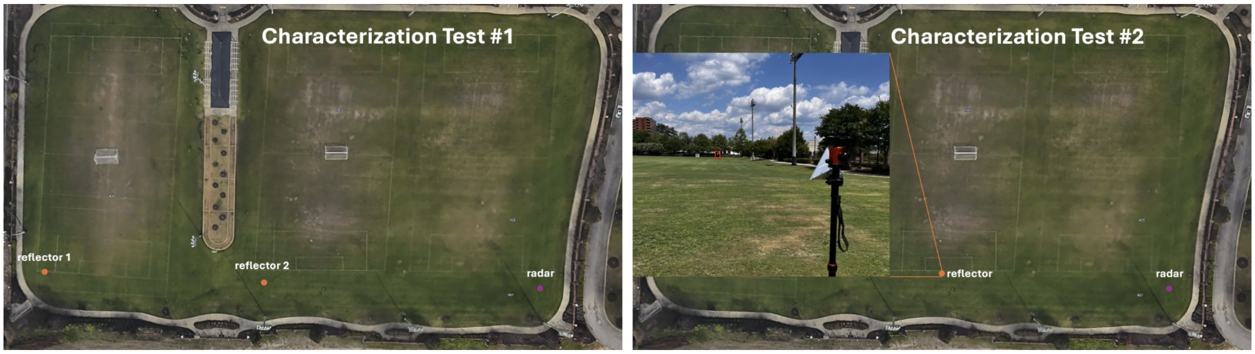

Figure 1: Overview of Strom Thurmond Field and Testing Setup

Figure 2: Results from Reflector at Strom Thurmond Field

A second campaign took place on a freshwater lake under the supervision of a junior operator. The first two captures served as pilot runs, applying the gain-scheduling concept outlined earlier to identify the three gains that would later feed the HDR stack. Range gates and pulse lengths were then varied to balance radial resolution against energy on target. Figure 3 displays PPI slices collected with the shortest, mid-length, and longest pulse options at a two-nautical-mile range setting.

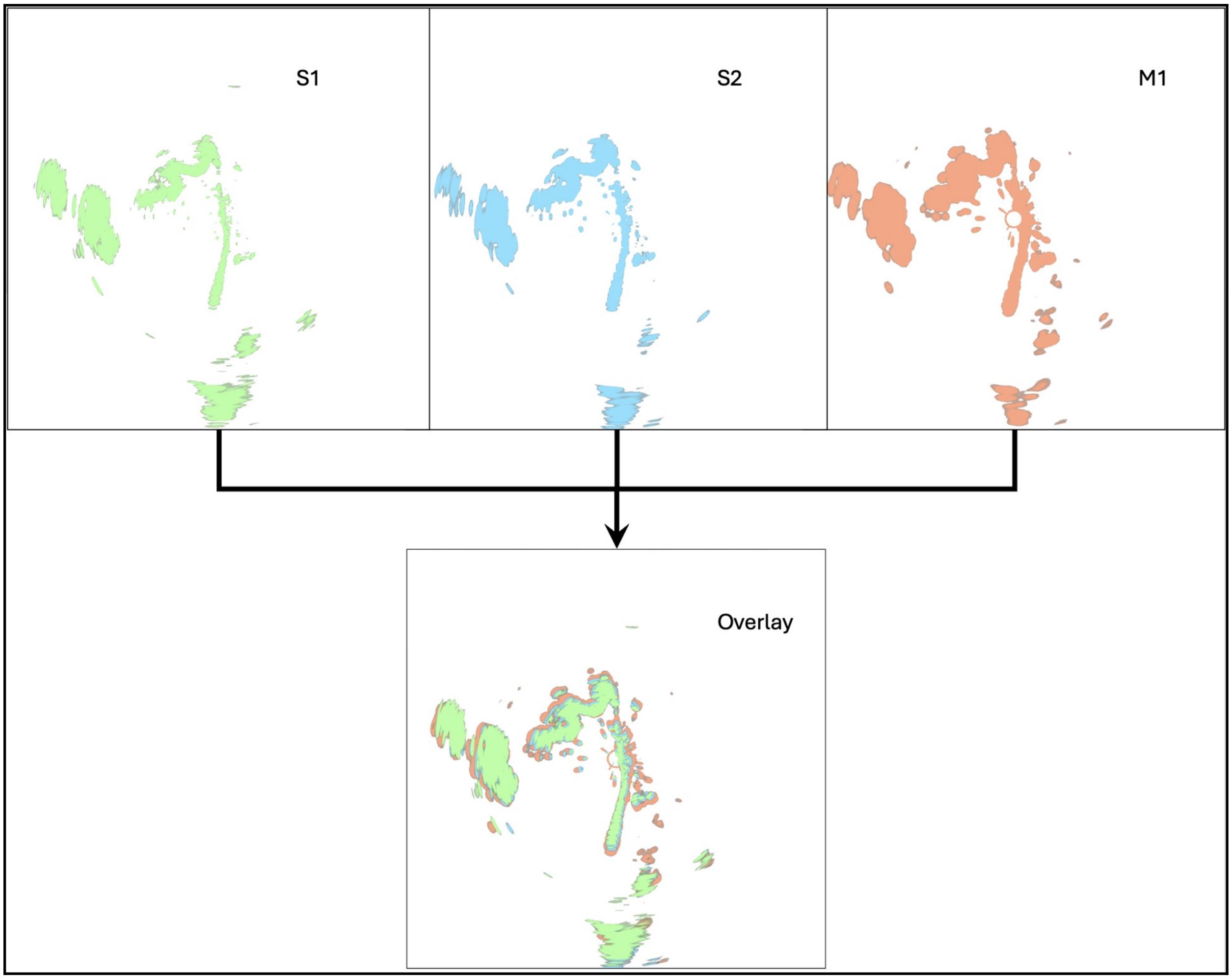

Figure 3: Pulse lengths S1, S2, M1 are shown at range 2, equivalent to 1 NM

Subsequent full collections used the short #2 pulse length and the HDR triplet (40, 50, 70); Figure 4 compares the 70-gain frame with its simulated counterpart, the simulation tuned with parameters drawn directly from the open-terrain characterization.

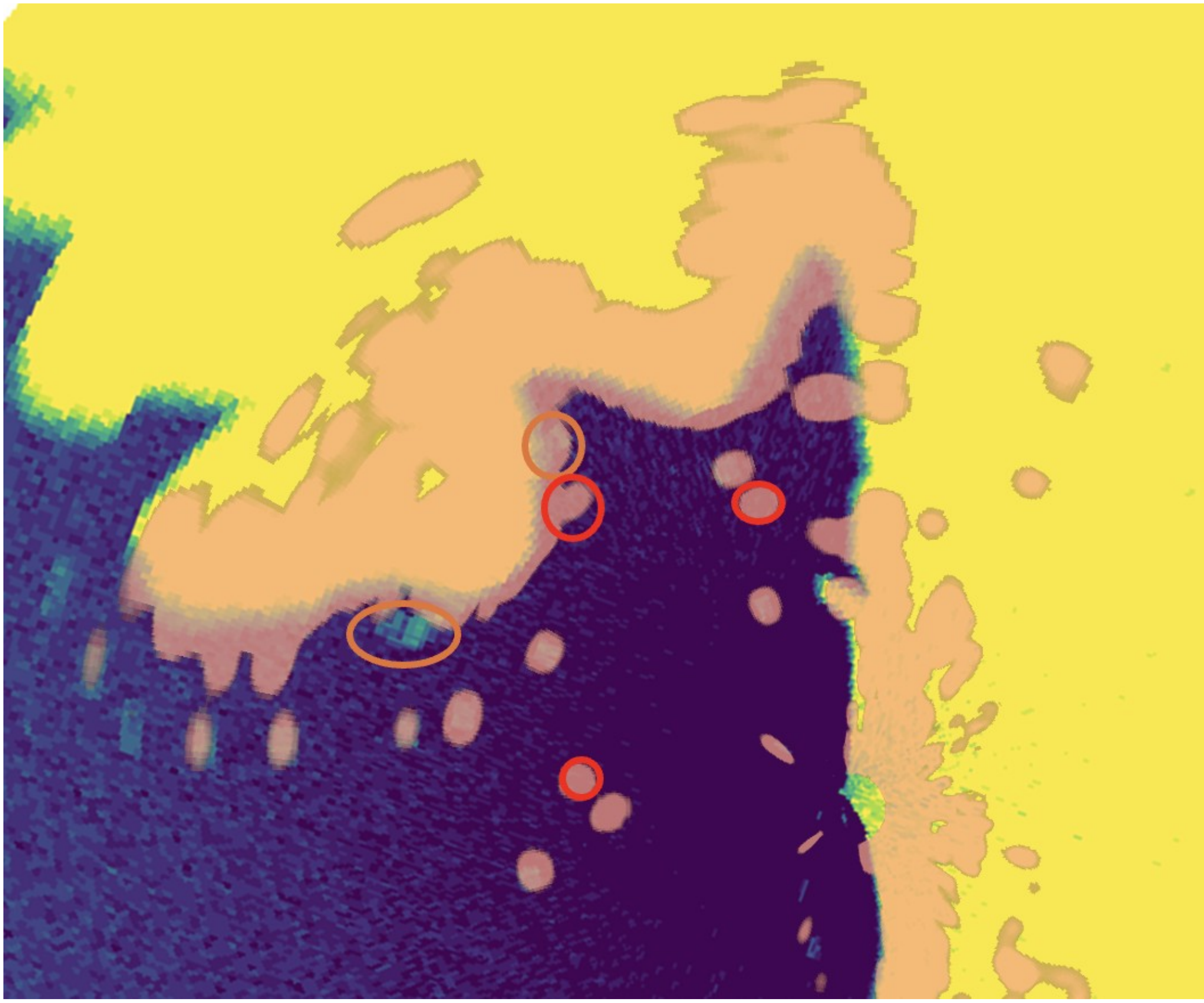

Figure 4: Gain 70 @ S2 radar data vs simulated data at similar parameters - with areas of interest highlighted

Part 3: High Dynamic Range Implementation

Lake-capture data, collected at the three gain levels identified during characterization, served as input to four candidate HDR algorithms. Three originated from the OpenCV ecosystem, Robertson, Mertens merge, and Debevec, while the fourth was an in-house extension that sought to recover sub-quantization detail from inherently low-bit-depth radar returns – implementation details are in July 2025 Report. Objective assessment on a downstream detection task revealed that the custom extension under-performed the established techniques, so it was set aside. Robertson's response curve, sigmoid-like in appearance, preserved both faint echoes and bright highlights without introducing halo artefacts; it was therefore promoted to production status.

HDR fusion lifted the probability of detection across all object classes and tightened track continuity by a clear margin; exact figures await an upcoming blind-label evaluation. Figure 5 presents hand-masked PPIs after HDR processing, demonstrating the gain in interpretability relative to single-gain imagery.

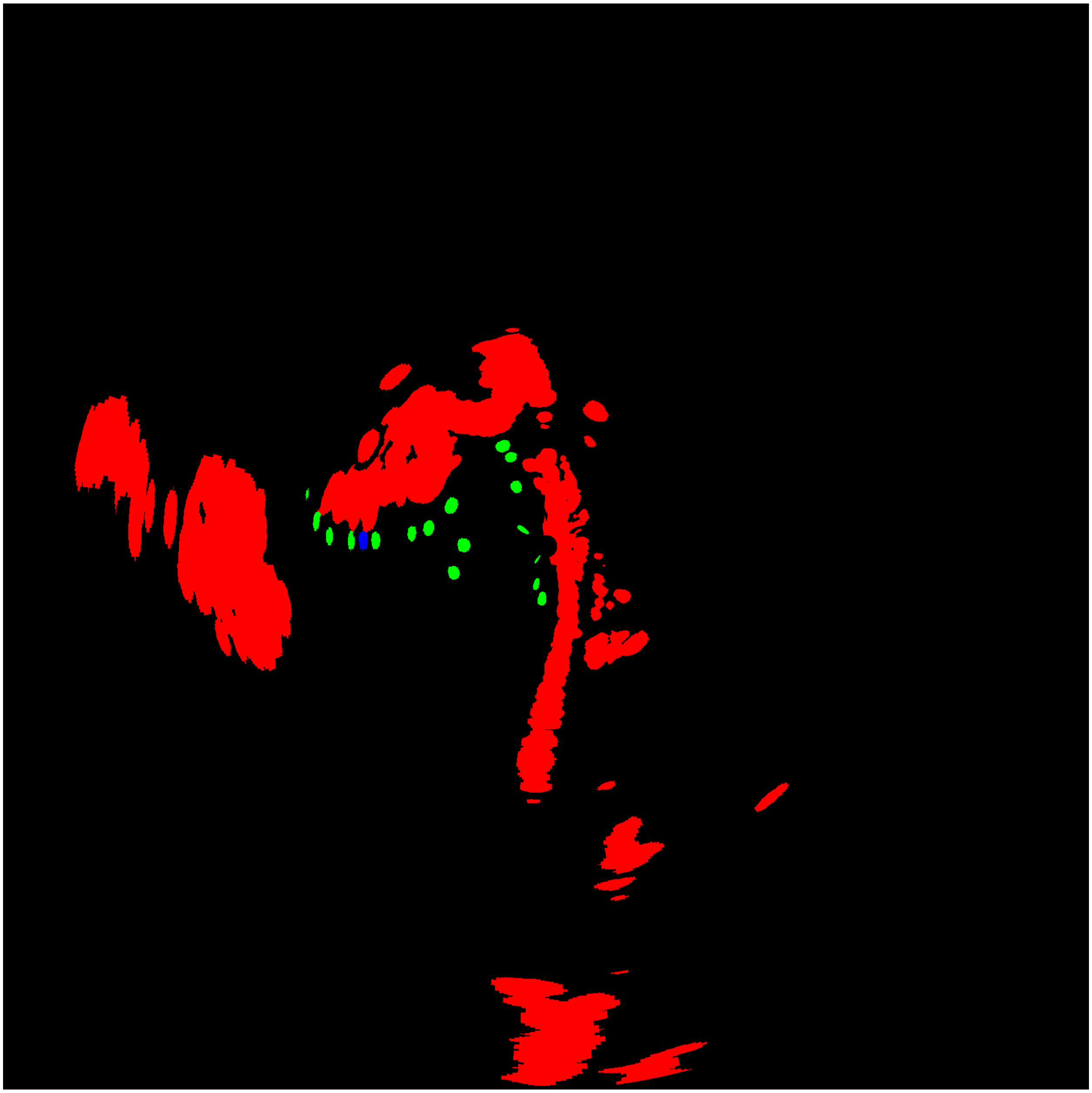

Figure 5: Masked HDR data; Red, Land; Green, Buoy; Blue, Boat

Part 4: Radar Simulation

A full end-to-end simulator generates range-azimuth power maps cell by cell. Each bin draws its amplitude from a user-selectable distribution, K, log-normal, Weibull, or standard Rayleigh, determined through Monte-Carlo draws whose statistics are anchored to the open-terrain characterization described in Part 2. The antenna pattern is supplied as an external table, allowing arbitrary gain roll-off and sidelobe structure to be reproduced exactly. Shorelines are retrieved automatically from public GIS layers and rasterized onto the same grid as the radar data, while buoy positions can be injected from a simple comma-separated list, simplifying direct bench-marking against live captures. Sensitivity-time-control curves are parameterized and interchangeable; the inverse-square law is the preset because it matches the radar equation best. Quantization and gain stages replicate the radar's front-end, producing 4-bit frames that are bit-for-bit comparable with recordings. Initial parameter seeds came from the corner-reflector trials; small offsets were then adjusted until simulated and real lake scenes aligned within two decibels at every range gate. Figure 6 overlays several STC choices on the same synthetic scene to illustrate the effect on near-range clutter suppression.

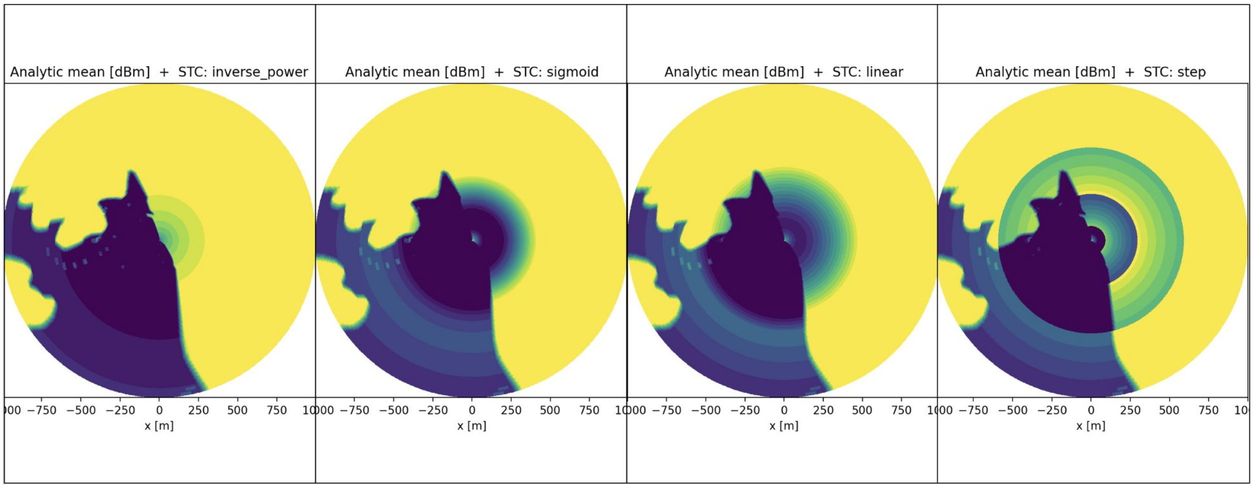

Figure 6: Comparison of STC Curves effect on simulated Lake Murray Data

Part 5: Object Tracking via Kalman

A single-scan detection stage feeds an extended Kalman filter that maintains position, velocity, and classification likelihood for every candidate target. Birth logic initiates a new track whenever consecutive detections exceed an adaptive signal-to-noise gate; conversely, tracks whose innovations fall below the gate for three scans are marked dormant rather than deleted, allowing the system to detect departures from piers or the re-emergence of temporarily occluded vessels. Static land contours are retained in a background layer and masked from display, yet remain accessible to the gating function so that slow-moving echoes adjacent to docks do not generate false alarms.

Qualitative comparison against hand-labelled truth (Figure 7) shows that the current pipeline captures almost every genuine boat but still admits a noticeable fraction of residual clutter, especially in high-gain regions. Quantitative refinement, tighter clutter covariance models and adaptive measurement gating, will be the focus of the next iteration.

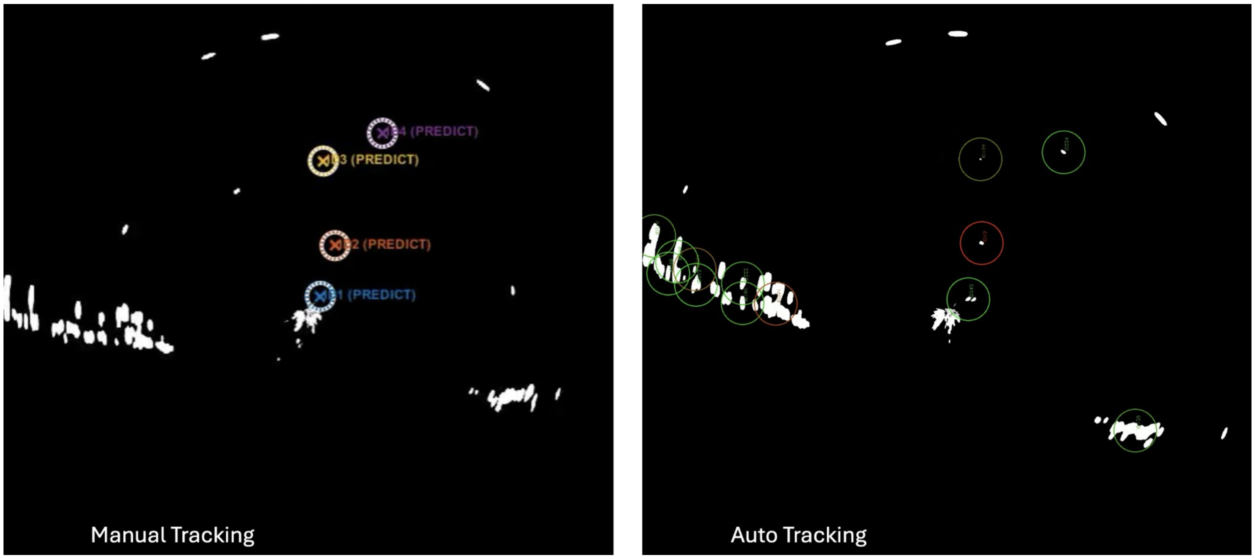

Figure 7: Human Labelled vs Auto-Labelled Data; note amount of clutter detected falsely

Part 6: Object Tracking via MAE - Preliminary

A temporal masked auto-encoder was trained exclusively on the HDR-enhanced lake imagery; no external scenes or labels were used. After self-supervised pre-training converged, a lightweight detection head was appended and fine-tuned on only three hand-drawn segmentation masks, demonstrating the feasibility of few-shot adaptation. Predicted masks remain noisy, small boats are fragmented and bright wave clutter occasionally survives, yet the mean intersection-over-union score is already several-fold higher than the baseline reported in July. Figure 8 compares the human reference masks against the auto-generated masks, highlighting both the progress so far and the regions that still demand refinement.

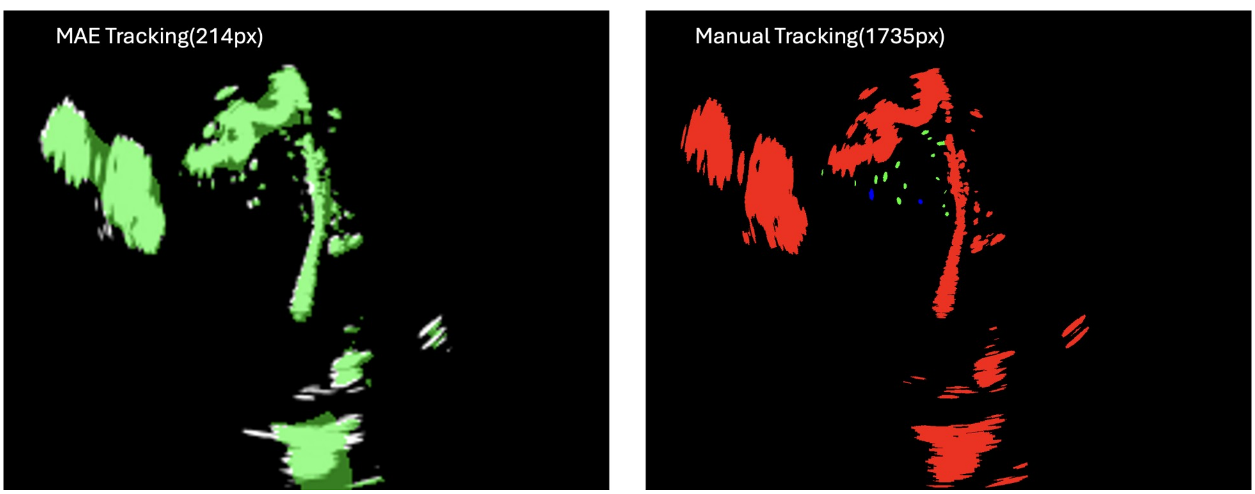

Figure 8: Human vs MAE masking results

Part 7 Future

Part 7a: Algorithms and Model Development
The Kalman tracker will receive an adaptive measurement-gating module and refined clutter-covariance estimates; in parallel, the masked auto-encoder will be retrained with longer temporal windows and a heavier masking ratio to improve few-shot segmentation fidelity. GPU access now available is expected to compress model iteration cycles from days to hours, while new command-line utilities will automate hyper-parameter sweeps and regression testing.

Part 7b: Experiments and Testing
Additional lake datasets will be collected under rain, fog, and choppy-surface conditions during the next reporting period, providing the variability needed to stress-test both tracking pipelines and to finalize quantitative performance benchmarks.